DE LA SALLE HOLY CROSS COLLEGE HIGH

Total Marks: 139

Time: 150 minutes

Examiner: Miss B Hannweg Moderator: Mrs D Webb

Name:

Instructions:

1. This question paper consists of 12 pages. Please ensure you have all pages.
2. There are two sections: Section A the answer booklet: 50 marks
   Section B: 89 marks
3. Section A is to be answered in the answer booklet given and section B on the folio paper provided.
4. Ensure your answer paper has your name on it.
5. It is in your own interest to write neatly and legibly.
6. **You may not use Tippex.**

Section	A					B				
Question number	1	2	3	4	5	6	7	8	9	10
Mark										
Total										

SECTION B

Instructions:

1. Section B is to be answered on the folio paper provided.
2. Ensure your answer papers have your name on them.
3. It is in your own interest to write neatly and legibly.
4. Leave a ruled margin on both the right and lefthand side of your page.
5. **You may not use Tippex.**

Question 6 [15]

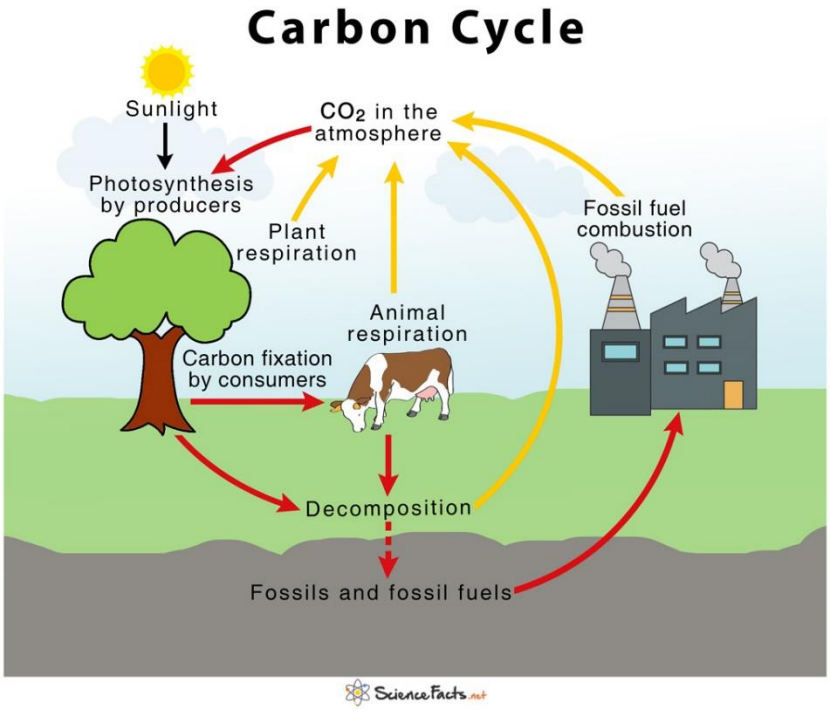
6.1. Phosphorus is needed to make ATP, DNA and phospholipids. Name the organelle or structure within the cell:

6.1.1. where ATP is manufactured. (1)

6.1.2. where DNA is found. (1)

6.1.3. that is made up of phospholipids. (1)

6.2. Below is a diagram of the carbon cycle.



Adapted from: https://socratic.org/questions/how-do-humans-interfere-in-the-carbon-cycle

6.2.1. Animals obtain their carbon in the same way that they obtain phosphorous. How do they obtain their carbon? (1)

6.2.2. Explain the role of the decomposers (bacteria and fungi) in the carbon cycle. (2)

6.2.3. Give two ways in which the role of PLANTS in this cycle compare to their Role in the WATER cycle. (2)

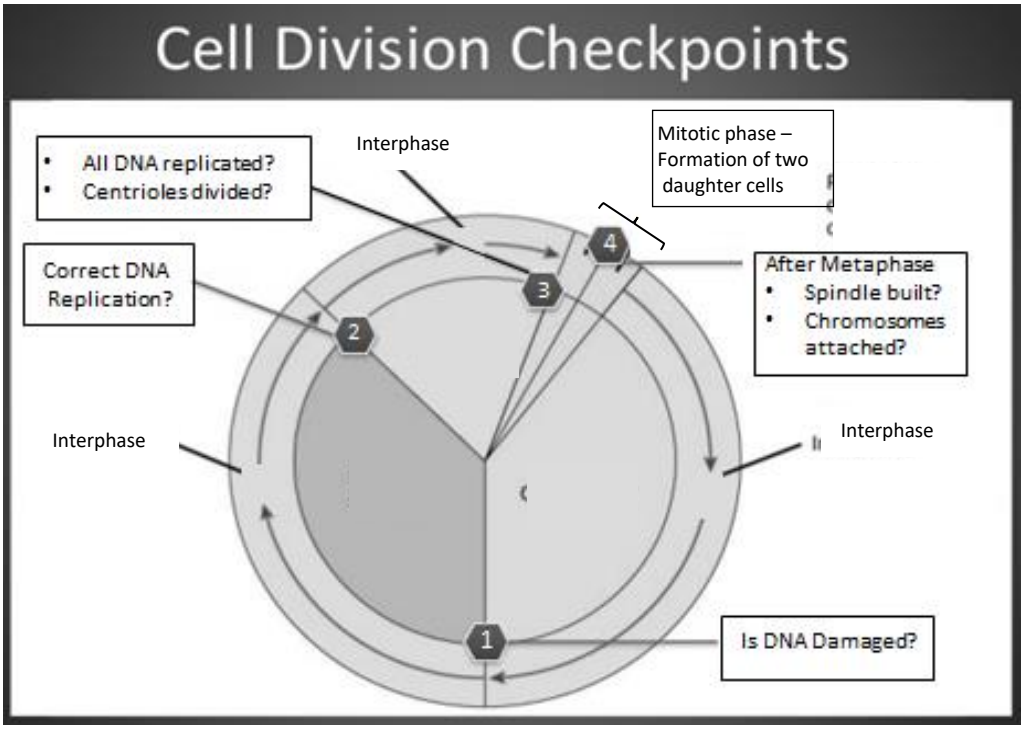
6.2.4. Explain why the importance of sunlight in the carbon cycle. (1)

6.2.5. What is the importance of nutrient cycles? (3)

6.2.6. Do humans have an impact on all four nutrient cycles? Explain. (3)

Question 7 **[9]**

7. The diagram below shows four different "checkpoints" at which a cell will "check" that mitosis is occurring correctly. These checkpoints help to control mitosis. Each checkpoint is represented by a number (1-4) in the diagram below.



Adapted from: https://quizlet.com/gb/264509129/biology-chapter-6-cell-cycle-checkpoints-diagram/

7.1. Name the longest phase of the cell cycle. (1)

7.2. Why is it essential that all the DNA is replicated prior to mitosis? (2)

7.3. What happens if there is an error in the duplication of DNA? (2)

7.4. Draw and label a diagram of a cell containing four chromosomes in the phase <u>following</u> metaphase. (4)

Question 8 **[12]**

8. Study the diagrams below and answer the questions that follow.

<table>
<tr><td>  </td><td>  </td></tr>
<tr><td>Source: https://plantlet.org/adaptation-of-plants/</td><td>Source: https://thepiquelab.com/blog/adaptations-of-a-pitcher-plant-the-flesh-eating-plant/</td></tr>
<tr><td>  </td><td>  </td></tr>
<tr><td>Source: https://www.edtechlens.com/blog/rainforest-kids-science-plants-control-rainforest</td><td>Source: https://www.sciencefocus.com/nature/five-astonishing-plant-adaptations/</td></tr>
</table>

8.1. Discuss with reference to the diagrams, why adaptions to leave may be of importance to plants. (4)

8.2. Choose one adaption that we have learnt about and explain how it will aid the plant in transpiration. (2)

8.3. What is the difference between the phloem and the Xylem? (2)

8.4. Plants may may use of two different phases to produce energy. Name these phases and describe the differences between them with reference to the reactants needed. (4)

Question 9 **[29]**

9. Read the article below and answer the questions which follow:

Asian elephants prefer habitats on the boundaries of protected areas

Adapted from: https://www.sciencedaily.com/releases/2022/10/221018091849.htm

New research, offering the most comprehensive analysis of Asian elephant movement and habitat preference to date, finds that elephants prefer habitats on the periphery of protected areas, rather than the areas themselves. The findings are published in the British Ecological Society's *Journal of Applied Ecology*.

An international team of researchers have analysed the movement and habitat preferences of 102 Asian elephants in Peninsular Malaysia and Borneo, recording over 600,000 GPS locations. They found that the majority of elephants spent more than half of their time outside of protected areas, preferring slightly disturbed forests and areas of regrowth. However, protected areas still played an important role, with the elephants' biggest preference being for areas within three kilometres of protected area boundaries.

It's thought that the preference for disturbed forest is related to food habits. Elephants like to eat grasses, bamboo, palms and fast-growing trees, which are common in disturbed environments, but relatively scarce under the canopy of old-growth forests.

Given their preference for habitats outside the protected areas, elephants will inevitably come into conflict with people. This highlights the importance of promoting human-elephant coexistence around protected areas.

The Sundaic region, where the research took place, is a global hotspot for biodiversity. However, it is estimated that only 50% of the region's original forest remains and less of 10% of it is formally protected. Asian elephants are endangered and live in highly fragmented landscapes in this region.

In the study, the researchers analysed the movement of 102 Asian elephants, recording over 60,000 GPS locations across the Malay peninsula and Borneo. The data was compiled from over a decade of fieldwork by three research groups. The researchers then compared this data with the locations of formally protected areas to see how much time elephants spent in these areas and the areas around them.

9.1. Identify two producers and a primary consumer mentioned. (3)

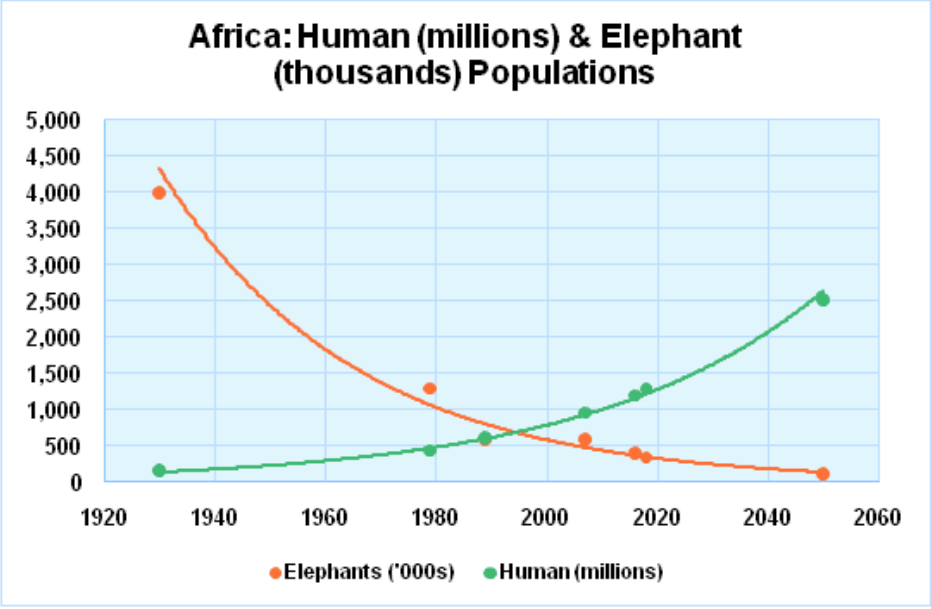
9.2. Why in your opinion, do you think there may inveribly be conflict between elephants and humans? (3)

9.3. Draw a food web to show FOUR trophic levels of a forest ecosystem. (4)

9.4. What does the term “Global hotspot for biodiversity” mean? Give an example of such an area in South Africa. (3)

9.5. The article refers to fragmented landscapes. In your opinion what does this mean? (2)

9.6. The graph below shows the results of an investigation that was conducted to determine the relationship between human and elephant population numers in Africa.



Source: https://wellbeingintl.org/the-last-elephants-2019-by-don-pinnock-colin-bell/

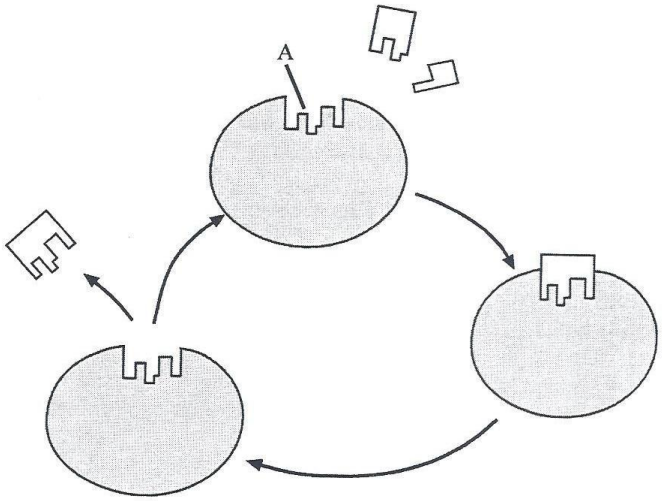
9.6.1. Explain how scientists may have collected this data. (2)

9.6.2. What would an independent and dependant variable be for this investigation? (2)

9.6.3. Looking at the years 1930, 1980, and 1990, construct a table which shows the differences in population numbers between the humans and elephants. (10)

Question 10 **[9]**

10. The following diagram represents the "lock and key" model which may be used to explain the way in which an enzyme works.



Adapted from: https://www.researchgate.net/figure/Lock-and-key-model-of-enzymatic-action_fig1_335505687

10.1. Name part A. (1)

10.2. Give TWO properties of enzymes that are SHOWN IN THE DIAGRAM. (2)

10.3. Describe how enzymes speed up the rate of a chemical reaction. (4)

10.4. Explain why increasing the temperature above the optimum causes the rate of areaction to decrease. (2)

Question 11 **[15]**

Look at the following sources provided and write a short essay (no longer than a page), as to why the procurement of nutrients are essential for the human body. Include in your writing the process that these nutrients go through when they are broken down.

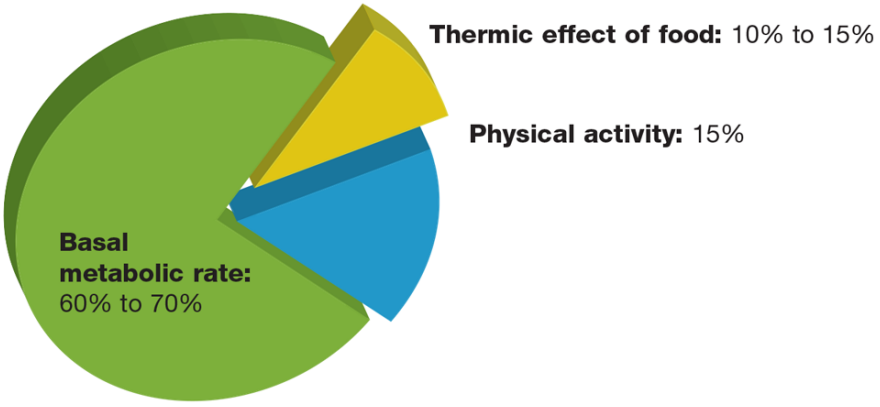
Source A

Eating disorders include anorexia nervosa, bulimia nervosa, binge eating disorder, and avoidant restrictive food intake disorder. They are second only to substance-abuse disorders in having the highest mental illness mortality rate. While mortality associated with anorexia nervosa has long been recognized, more recently disorders previously categorized as "eating disorder not otherwise specified," and now categorized as "other specified feeding or eating disorders," are reported to be most prevalent and to have the highest mortality of any category of eating disorder.

Source: https://www.psychiatrictimes.com/view/understanding-nutritional-needs-patients-eating-disorders-implications-psychiatrists

Source B

Figure. Major components of energy expenditure



Source: https://www.psychiatrictimes.com/view-psychiatrists

Source C

6 Essential Nutrients and Why Your Body Needs Them

- Protein.
- Carbs.
- Fats.
- Vitamins.
- Minerals.
- Water.

Source: https://www.healthline.com/health/food-nutrition/six-essential-nutrients

Source D



Source: https://www.istockphoto.com/search/2/image?mediatype=illustration&phrase=nutrition+superhero

THE END!